

Project Summary - Indigo Wellness Inventory Control

1. The Problem

Indigo Wellness Group operates two wellness centers in Stamford and Westport, CT. Inventory was managed manually and often overlooked, creating some recurring operational failures. In addition, there are concerns about handling sensitive information regarding HIPPA regulations.

- **Running out of stock** — no visibility on inventory on hands
- **Unable to locate inventory** — no knowledge regarding inventory stock in both locations, causing time lost searching and missed sales.
- **Untracked transfers** — products moved between sites to cover committed bookings with no record of who moved what, when, or why.

2. The Solution

A single-file, browser-based inventory system built for Indigo Wellness. No installation required, just open the file, enter data daily, and everything saves locally in your browser, even without an internet connection (for most tasks). Key capabilities delivered:

- **Per-location stock tracking** across Stamford and Westport with live dashboard, status alerts (OK / Low / Critical), and donut charts.
- **Smart reorder points** auto-calculated from batch size × average supplier lead time. Overdue shipments auto-flip to Delayed.
- **Shipment management** with AI extraction — drop a PDF or Excel order and the system reads supplier, ETA, location, and products automatically.
- **Inventory Update (Purchase/Sales)** with AI extraction, manual entry, duplicate detection (content hash + PO number), and an editable review table before committing.
- **Transfer log** with reason tracking (committed sale, stock balancing, event) and full audit trail per location.
- **Trends & Forecast** — 7-day usage rates, per-location Stamford vs Westport consumption chart, stockout forecast, and reorder recommendations.

3. AI Tools, Technologies & Techniques

AI Use Cases

Use case	What AI does	Technique
Order extraction	Reads PDF/Excel/text → extracts supplier, ETA, products, quantities	Strict JSON schema prompt; output reviewed in editable table before committing
Location detection	Identifies Stamford vs Westport from address/ZIP in document	Address-aware prompt + client-side regex fallback (25 Sylvan Rd / 66 Glenbrook Rd)
Category assignment	Tags each product as Health, Massage, Retail, or General	Category list injected into system prompt; user can override per item

Technology Stack

Technology	Purpose
HTML / CSS / JavaScript	Single-file application — no framework, no build step, offline-capable
Anthropic Claude API	Direct browser-to-API extraction and category detection
SheetJS (XLSX)	Client-side Excel/CSV parsing before sending to AI
Chart.js + Canvas API	Dashboard donut charts and trend bar charts
localStorage	Browser-native persistence — items, shipments, transfers, history, fingerprints

AI Techniques

- **Structured output prompting** — all API calls return strict JSON; never free-form text. Parsed and validated client-side.
- **Human-in-the-loop review** — AI output populates an editable table; users correct, add, or remove rows before any data is committed.
- **Two-layer duplicate detection** — FNV-1a full-content hash catches re-uploads; regex PO/invoice number extraction catches the same order in different files.
- **Graceful degradation** — if per-location history is missing, falls back to proportional estimates from overall usage rate, and is flagged with an "est." badge.

AI Model & Integration

Anthropic Claude API — claude-sonnet-4-6 — called directly from the browser using anthropic-dangerous-direct-browser-access. No backend required.

4. Outcome

The system gives Indigo Wellness a single source of truth for inventory across both locations, accessible from any browser with no IT infrastructure. AI eliminates manual re-keying of order data, reducing processing time from minutes to seconds. The transfer log and per-location stock views directly address the core pain points, while stockout forecasting shifts the team from reactive restocking to proactive planning.